

CSE 4412: Introduction to Quantum Computing

Homework 1 — Chapter 2 Exercises

Exercise 2.2

Statement. Let $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Compute $\langle +|+\rangle$, $\langle +|-\rangle$, and $\langle -|-\rangle$. What can you say about these two vectors? You may use the fact that $|0\rangle$ and $|1\rangle$ are orthonormal.

Hint. Expand each bracket using bilinearity of the inner product (distribute over the $\pm$ sums), then substitute $\langle 0|0\rangle = \langle 1|1\rangle = 1$ and $\langle 0|1\rangle = \langle 1|0\rangle = 0$. Once you have all three numbers, think about what “normalized” and “orthogonal” mean (Definitions in Section 2.1.2) and what it implies when two normalized, mutually orthogonal vectors span the whole space.

Your answer:

Expand each inner product using bilinearity, then substitute $\langle 0|0\rangle = \langle 1|1\rangle = 1$ and $\langle 0|1\rangle = \langle 1|0\rangle = 0$.

$$\langle +|+\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} (\langle 0|0\rangle + \langle 0|1\rangle + \langle 1|0\rangle + \langle 1|1\rangle) = \frac{1}{2}(1 + 0 + 0 + 1) = 1$$

$$\langle +|-\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} (\langle 0|0\rangle - \langle 0|1\rangle + \langle 1|0\rangle - \langle 1|1\rangle) = \frac{1}{2}(1 - 0 + 0 - 1) = 0$$

$$\langle -|-\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} (\langle 0|0\rangle - \langle 0|1\rangle - \langle 1|0\rangle + \langle 1|1\rangle) = \frac{1}{2}(1 - 0 - 0 + 1) = 1$$

Since $\langle +|+\rangle = \langle -|-\rangle = 1$, both $|+\rangle$ and $|-\rangle$ are normalized (unit vectors), and since $\langle +|-\rangle = 0$, they are orthogonal to each other. Two normalized, mutually orthogonal vectors in a 2-dimensional space form an orthonormal basis, so $\{|+\rangle, |-\rangle\}$ is an alternative orthonormal basis for the qubit state space (alongside $\{|0\rangle, |1\rangle\}$).

Exercise 2.4

Statement. Consider the following three questions:

1. Consider the $|+\rangle$ state. What is the probability of observing $|1\rangle$ after a measurement?
2. Consider $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. What is the probability of observing $|1\rangle$ after measuring that state?
3. Consider the state $|1\rangle$. What is the probability of observing $|1\rangle$ after measurement?

Hint. For a state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, the rule from the chapter is $\Pr(|1\rangle) = |\beta|^2$, where β is whatever is multiplying $|1\rangle$ once the state is written in the $\{|0\rangle, |1\rangle\}$ basis. Identify β in each of the three given states (watch the sign in $|-\rangle$ — remember you square the *magnitude*).

Your answer:

For a state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $\Pr(|1\rangle) = |\beta|^2$.

1. $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$, so $\beta = \frac{1}{\sqrt{2}}$ and

$$\Pr(|1\rangle) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}.$$

2. $|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$, so $\beta = -\frac{1}{\sqrt{2}}$. Squaring the *magnitude* discards the sign, so

$$\Pr(|1\rangle) = \left| -\frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2},$$

the same probability as for $|+\rangle$, even though the two states are orthogonal (Exercise 2.2) — the relative phase (sign) between α and β does not affect the measurement probabilities in the computational basis.

3. $|1\rangle = 0 \cdot |0\rangle + 1 \cdot |1\rangle$, so $\beta = 1$ and

$$\Pr(|1\rangle) = |1|^2 = 1,$$

i.e. measuring $|1\rangle$ always yields “1” with certainty, as expected since the state already is a computational basis state.

Exercise 2.5

Statement. Imagine you are given a qubit $|\psi\rangle$ but not told anything about it (i.e., you do not know its probability amplitudes). If you perform a measurement on the state and the outcome is “0,” what does this tell you about the state? Can you learn the original probability amplitudes? What if you are given as many copies of $|\psi\rangle$ as you want — what can be said then?

Hint. Separate the question into two parts. (a) Single measurement: think about *state collapse* — what do you know about the state right after the measurement, versus what you knew about it beforehand? Can one classical bit of outcome data pin down two continuous amplitudes α, β ? (b) Many copies: if you measured each copy and tallied the fraction of “0” vs. “1” outcomes, what would that fraction converge to as the number of copies grows (law of large numbers)? Does that let you recover α and β exactly, or only something about their magnitudes? Also recall the no-cloning theorem (mentioned in the text, Section 2.5.4) — why can’t you manufacture the extra copies yourself?

Your answer:

Exercise 2.6

Statement. Let $p \in [0, 1]$ be a real number and consider the state

$$|\psi\rangle = \sqrt{p} |0\rangle + \sqrt{1-p} |1\rangle.$$

Answer the following: (1) probability of observing $|0\rangle$ for a given p ; (2) probability of observing $|-\rangle$ for a given p ; (3)–(4) graph both probability distributions as functions of p .

Hint. For (1), same rule as Exercise 2.4, just applied to $|0\rangle$ instead of $|1\rangle$. For (2), you can't just read off a coefficient because $|-\rangle$ isn't one of the two basis vectors $|\psi\rangle$ is written in terms of — instead use the inner-product rule $\Pr(|-\rangle) = |\langle -|\psi\rangle|^2$ from Section 2.3.1 (Equation 2.36), and expand $\langle -|\psi\rangle$ using linearity and the known values of $\langle 0|0\rangle, \langle 0|1\rangle, \langle 1|0\rangle, \langle 1|1\rangle$. For (3)–(4), once you have closed-form expressions in terms of p , you can plot them with pgfplots (`\usepackage{pgfplots}`) or by hand on $[0, 1]$ — try a few sample points ($p = 0, \frac{1}{2}, 1$) first to sanity-check the shape before you commit to plotting code.

Your answer:

Exercise 2.8

Statement. Let $|\psi\rangle$ be some d -dimensional state and let $\mathcal{B} = \{|b_0\rangle, \dots, |b_{d-1}\rangle\}$ be any orthonormal basis. Show that $|\psi\rangle$ may be written as

$$|\psi\rangle = \sum_{j=0}^{d-1} \langle b_j|\psi\rangle |b_j\rangle.$$

Hint. This is essentially asking you to re-derive the *completeness relation* argument from Section 2.1.3 (Equation 2.22–2.23). Start from the fact that $\sum_j |b_j\rangle \langle b_j| = I$ for any orthonormal basis, apply both sides to $|\psi\rangle$, and note that $|b_j\rangle \langle b_j|\psi\rangle$ can be regrouped as a scalar $\langle b_j|\psi\rangle$ times the vector $|b_j\rangle$.

Your answer: